

Supplementary Material: Mathematical Proofs, Algorithmic Invariants, and Extended Experimental Details for FlowOpt

Anonymous Authors

APPENDIX A

FORMAL MATHEMATICAL PROOFS

A.1 Proof of Theorem 1: Closed-Form Bayes-Optimal Velocity Field

Theorem 1. Let $p_0(x)$ and $p_1(x_1)$ denote probability measures on $\mathbb{R}^d$, and let $x_t = \psi_t(x_0, x_1) = (1-t)x_0 + tx_1$ denote the linear optimal transport displacement interpolant with conditional velocity $u_t(x_t | x_0, x_1) = x_1 - x_0$. The velocity field $v^*(x, t)$ minimizing the continuous Flow Matching objective:

$$\mathcal{L}_{FM}(v) = \mathbb{E}_{t \sim \mathcal{U}[0,1], (x_0, x_1) \sim \pi} [\|v(x_t, t) - (x_1 - x_0)\|^2] \quad (1)$$

is uniquely given by the conditional expectation:

$$v^*(x, t) = \mathbb{E}_{(x_0, x_1) \sim \pi} [x_1 - x_0 | x_t = x] \quad (2)$$

Furthermore, for a discrete empirical ensemble of particle pairs $\{(x_0^{(i)}, \hat{x}_1^{(i)})\}_{i=1}^N$ with local Gaussian kernel density smoothing $p_t(x) \approx \frac{1}{N} \sum_{i=1}^N K_h(x, x_t^{(i)})$, the estimator takes the closed form:

$$\hat{v}^*(x, t) = \sum_{i=1}^N \frac{K_h(x, x_t^{(i)})}{\sum_{j=1}^N K_h(x, x_t^{(j)}) + \epsilon} (\hat{x}_1^{(i)} - x_0^{(i)}) \quad (3)$$

Proof. By Fubini's theorem and the law of total expectation, we expand the regression objective for any fixed time t :

$$\mathcal{L}_t(v) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^{2d}} \|v(x, t) - (x_1 - x_0)\|^2 \times p(x_0, x_1 | x_t = x) p_t(x) dx_0 dx_1 dx \quad (4)$$

Since $p_t(x) \geq 0$, minimizing the total integral with respect to $v(x, t)$ is equivalent to pointwise minimization of the inner expectation:

$$v^*(x, t) = \arg \min_{v \in \mathbb{R}^d} \mathbb{E}_{(x_0, x_1) \sim \pi} [\|v - (x_1 - x_0)\|^2 | x_t = x] \quad (5)$$

Expanding the quadratic norm:

$$\|v - (x_1 - x_0)\|^2 = \|v\|^2 - 2\langle v, x_1 - x_0 \rangle + \|x_1 - x_0\|^2 \quad (6)$$

Taking the gradient with respect to v and setting it to zero:

$$\begin{aligned} \nabla_v \mathbb{E} [\|v - (x_1 - x_0)\|^2 | x_t = x] \\ = 2v - 2\mathbb{E} [x_1 - x_0 | x_t = x] = 0 \end{aligned} \quad (7)$$

yielding the Bayes-optimal solution:

$$v^*(x, t) = \mathbb{E}[x_1 - x_0 | x_t = x] \quad (8)$$

Now, consider an empirical joint sample $\{(x_0^{(i)}, \hat{x}_1^{(i)})\}_{i=1}^N$ with intermediate points $x_t^{(i)} = (1-t)x_0^{(i)} + t\hat{x}_1^{(i)}$. By non-parametric kernel density estimation with kernel $K_h(\cdot, \cdot)$:

$$p(x, u) \approx \frac{1}{N} \sum_{i=1}^N K_h(x, x_t^{(i)}) \delta(u - u^{(i)}) \quad (9)$$

where $u^{(i)} = \hat{x}_1^{(i)} - x_0^{(i)}$. Applying the classic Nadaraya-Watson kernel regression formula:

$$\hat{v}^*(x, t) = \frac{\int u p(x, u) du}{\int p(x, u) du} = \frac{\sum_{i=1}^N K_h(x, x_t^{(i)}) u^{(i)}}{\sum_{j=1}^N K_h(x, x_t^{(j)}) + \epsilon} \quad (10)$$

which concludes the proof. $\square$

A.2 Proof of Theorem 2: Kinetic Action Minimization via Optimal Transport

Theorem 2. Let $\mathcal{C}(p_0, p_1)$ denote the set of time-dependent probability paths $(p_t, v_t)_{t \in [0,1]}$ satisfying the continuity equation $\partial_t p_t + \nabla \cdot (p_t v_t) = 0$ with boundary conditions $p_{t=0} = p_0$ and $p_{t=1} = p_1$. The 2-Wasserstein Benamou-Brenier dynamic formulation states:

$$W_2^2(p_0, p_1) = \inf_{(p_t, v_t) \in \mathcal{C}(p_0, p_1)} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 p_t(x) dx dt \quad (11)$$

The linear displacement interpolant coupled via Monge-Kantorovich transport π^* achieves the global minimum of the total kinetic action $\mathcal{S}[v] = \frac{1}{2} \int_0^1 \mathbb{E}_{p_t} [\|v_t\|^2] dt$.

Proof. By Jensen's inequality applied to the convex kinetic energy function $\phi(v) = \|v\|^2$:

$$\int_0^1 \|v(x_t, t)\|^2 dt \geq \left\| \int_0^1 v(x_t, t) dt \right\|^2 = \|x_1 - x_0\|^2 \quad (12)$$

Equality holds if and only if $v(x_t, t)$ is constant in time along each characteristic trajectory:

$$\frac{dx_t}{dt} = x_1 - x_0 \implies x_t = (1-t)x_0 + tx_1 \quad (13)$$

Integrating over the optimal coupling $\pi^*(x_0, x_1)$:

$$\mathcal{S}[v^*] = \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x_1 - x_0\|^2 d\pi^*(x_0, x_1) = \frac{1}{2} W_2^2(p_0, p_1) \quad (14)$$

Any independent coupling $\pi_{\text{indep}} = p_0 \otimes p_1$ yields $\mathbb{E}_{\pi_{\text{indep}}}[\|x_1 - x_0\|^2] \geq W_2^2(p_0, p_1)$, with path crossings causing non-zero trajectory curvature and turbulent kinetic energy dissipation. Entropic Optimal Transport regularizes π^* with entropy parameter ε_{OT} , ensuring smooth, deterministic, geodesic velocity fields. $\square$

A.3 Proof of Theorem 3: Variance Calibration of Step-Size Adaptation

Theorem 3. Let $z_1, \dots, z_\mu \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_D)$ and let $w \in \Delta^{\mu-1}$ be positive normalized weights with effective selection mass $\mu_{\text{eff}} = 1/\sum_{i=1}^\mu w_i^2$. Under the null hypothesis of pure random drift (i.e., objective function $f(x) \equiv \text{const}$), the flow displacement vector:

$$z_{\text{flow}} = \sqrt{\mu_{\text{eff}}} \sum_{i=1}^\mu w_i z_i \quad (15)$$

satisfies:

$$z_{\text{flow}} \sim \mathcal{N}(0, I_D) \quad (16)$$

Consequently, the expected norm of the cumulative flow path p_σ in the stationary state is:

$$\begin{aligned} \mathbb{E}[\|p_\sigma\|] &= \chi_D \equiv \sqrt{2} \frac{\Gamma((D+1)/2)}{\Gamma(D/2)} \\ &= \sqrt{D} \left(1 - \frac{1}{4D} + \frac{1}{21D^2} + \mathcal{O}(D^{-3})\right) \end{aligned} \quad (17)$$

guaranteeing an unbiased step-size adaptation drift:

$$\mathbb{E} \left[\ln \frac{\sigma_{t+1}}{\sigma_t} \mid \text{null hypothesis} \right] = 0 \quad (18)$$

Proof. Since each $z_i \sim \mathcal{N}(0, I_D)$ are independent, the linear combination $Y = \sum_{i=1}^\mu w_i z_i$ is a Gaussian random vector with mean $\mathbb{E}[Y] = 0$ and covariance:

$$\text{Cov}(Y) = \sum_{i=1}^\mu w_i^2 \text{Cov}(z_i) = \left(\sum_{i=1}^\mu w_i^2 \right) I_D = \frac{1}{\mu_{\text{eff}}} I_D \quad (19)$$

Multiplying by $\sqrt{\mu_{\text{eff}}}$:

$$\text{Cov}(z_{\text{flow}}) = \mu_{\text{eff}} \cdot \text{Cov}(Y) = I_D \quad (20)$$

Thus $z_{\text{flow}} \sim \mathcal{N}(0, I_D)$. The cumulative flow path evolves as:

$$p_\sigma^{(t+1)} = (1 - c_\sigma) p_\sigma^{(t)} + \sqrt{c_\sigma(2 - c_\sigma)} z_{\text{flow}}^{(t)} \quad (21)$$

In the stationary limit $t \rightarrow \infty$, the stationary covariance Σ_{p_σ} satisfies:

$$\Sigma_{p_\sigma} = (1 - c_\sigma)^2 \Sigma_{p_\sigma} + c_\sigma(2 - c_\sigma) I_D \implies \Sigma_{p_\sigma} = I_D \quad (22)$$

Hence $p_\sigma \sim \mathcal{N}(0, I_D)$, whose norm $\|p_\sigma\|$ follows the Chi distribution with D degrees of freedom. The expectation of a Chi- D variable is:

$$\mathbb{E}[\|p_\sigma\|] = \sqrt{2} \frac{\Gamma((D+1)/2)}{\Gamma(D/2)} \equiv \chi_D \quad (23)$$

Applying Stirling's asymptotic expansion of the Gamma ratio:

$$\frac{\Gamma(x+1/2)}{\Gamma(x)} = \sqrt{x} \left(1 - \frac{1}{8x} + \frac{1}{128x^2} + \dots\right) \quad (24)$$

Setting $x = D/2$:

$$\chi_D = \sqrt{2} \sqrt{\frac{D}{2}} \left(1 - \frac{1}{4D} + \frac{1}{21D^2} + \mathcal{O}(D^{-3})\right) \quad (25)$$

Under the log-step-size update rule:

$$\ln \sigma_{t+1} - \ln \sigma_t = \frac{c_\sigma}{d_\sigma} \left(\frac{\|p_\sigma\|}{\chi_D} - 1 \right) \quad (26)$$

Taking the expectation gives $\mathbb{E}[\|p_\sigma\|/\chi_D - 1] = 1 - 1 = 0$, proving that step-size drift is exactly zero under the null hypothesis. Without the $\sqrt{\mu_{\text{eff}}}$ prefactor, $\text{Cov}(Y) = \frac{1}{\mu_{\text{eff}}} I_D$, which forces $\mathbb{E}[\|p_\sigma\|] = \chi_D/\sqrt{\mu_{\text{eff}}} \approx 0.515\chi_D$. This would result in a permanent negative drift $\frac{c_\sigma}{d_\sigma}(0.515 - 1) \approx -0.07$, decaying σ_t by 7% each generation regardless of performance. $\square$

APPENDIX B

BENCHMARK FUNCTION DEFINITIONS

Table 1 enumerates the mathematical formulations, standard domain bounds, and global optima for all eight evaluated benchmark problems in $D = 10$ dimensions.

APPENDIX C

BASELINE ALGORITHMS IMPLEMENTATION AND HYPERPARAMETERS

All baseline algorithms were executed under identical conditions: standard popsize $N = 30$, budget $T = 200$ iterations, and 5 random seeds (42, 43, 44, 45, 46).

- 1) **CMA-ES (Covariance Matrix Adaptation Evolution Strategy)**: Canonical Hansen (2001) implementation with cumulative path length adaptation, rank-1 and rank- μ covariance updates.
- 2) **DE (Differential Evolution)**: Standard DE/rand/1/bin strategy with mutation scaling factor $F = 0.8$ and binomial crossover probability $CR = 0.9$.
- 3) **PSO (Particle Swarm Optimization)**: Standard velocity clamping with inertia weight $w = 0.729$, cognitive acceleration $c_1 = 1.494$, and social acceleration $c_2 = 1.494$.
- 4) **CBO (Consensus-Based Optimization)**: Analytical CBO formulation (Carrillo et al., 2018) with drift rate $\lambda = 1.0$, isotropic diffusion coefficient $\sigma_B = 0.4$, and Gibbs consensus weight temperature $\alpha = 50.0$.
- 5) **CEM (Cross-Entropy Method)**: Elite selection ratio $\rho = 0.2$, smoothing factor $\alpha = 0.7$ for Gaussian mean and variance updates.

TABLE 1
Mathematical Definitions of Benchmark Objective Functions ($D = 10$).

Name	Formulation	Domain Ω	$f(x^*)$
Sphere	$f(x) = \sum_{i=1}^D x_i^2$	$[-5.12, 5.12]^D$	0.0 at $x^* = \mathbf{0}$
Rosenbrock	$f(x) = \sum_{i=1}^{D-1} [100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2]$	$[-5.0, 10.0]^D$	0.0 at $x^* = \mathbf{1}$
Rastrigin	$f(x) = 10D + \sum_{i=1}^D [x_i^2 - 10 \cos(2\pi x_i)]$	$[-5.12, 5.12]^D$	0.0 at $x^* = \mathbf{0}$
Ackley	$f(x) = -20 \exp \left(-0.2 \sqrt{\frac{1}{D} \sum_{i=1}^D x_i^2} \right) - \exp \left(\frac{1}{D} \sum_{i=1}^D \cos(2\pi x_i) \right) + 20 + e$	$[-32.768, 32.768]^D$	0.0 at $x^* = \mathbf{0}$
Griewank	$f(x) = 1 + \frac{1}{4000} \sum_{i=1}^D x_i^2 - \prod_{i=1}^D \cos(x_i / \sqrt{i})$	$[-600.0, 600.0]^D$	0.0 at $x^* = \mathbf{0}$
Schwefel	$f(x) = 418.9829D - \sum_{i=1}^D x_i \sin(\sqrt{ x_i })$	$[-500.0, 500.0]^D$	0.0 at $x^* = 420.9687 \cdot \mathbf{1}$
Levy	$f(x) = \sin^2(\pi w_1) + \sum_{i=1}^{D-1} (w_i - 1)^2 [1 + 10 \sin^2(\pi w_i + 1)] + (w_D - 1)^2 [1 + \sin^2(2\pi w_D)]$	$[-10.0, 10.0]^D$	0.0 at $x^* = \mathbf{1}$
Lennard-Jones	$V(r) = 4\epsilon \sum_{i < j} [(\sigma/r_{ij})^{12} - (\sigma/r_{ij})^6], N_{\text{atom}} = 3, D = 9 \text{ coords}$	$[-2.0, 2.0]^D$	-3.0 (equilateral triangle)